D833 Task 1 Topic Approval Form

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The purpose of this document is to help you clearly explain your capstone topic, including the cybersecurity problem technical solution, stakeholders, and implementation plan. Without clearly addressing each of these areas, you will not have a complete and realistic overview of your project and your instructor cannot accurately assess whether your project will be viable for this course.

Complete this form and email it to your instructor for approval. Once approved, you will receive a signed PDF that you can upload as part of Task 1.

You can submit your signed form to your instructor as a .pdf, .doc, or .docx file. The form must include an actual signature – either digital or handwritten. Typing your name into the field does not meet the signature requirement.

It is the policy of Western Governors University (WGU) that student capstone projects should not be based on or include, without authorization, restricted information. Restricted information is any proprietary or classified information or material belonging to your employer or any other third party. You acknowledge that you will not use restricted information in your capstone project without obtaining the third party's permission by using the "Restricted Information Authorization Form" in the Supporting Documents section of Task 1.

Cybersecurity Problem

Context: *Provide an overview of the current security environment and the context for the identified risk or vulnerability.*

For my capstone project, I am focusing on a fictional organization named VirtuTrain Solutions. VirtuTrain is a privately held software development company with 450 employees operating in the corporate education sector, providing custom VR training simulations to large enterprise clients. The company is currently rolling out 150 Meta Quest Pro Virtual Reality headsets to its remote developers and QA testers to facilitate immersive training and remote team collaboration. While researching this technology, I found that these devices need to continuously collect and process a massive amount of sensitive telemetry data to function properly. This includes mapping the user's physical environment, recording voices, and tracking biometrics like eye and facial movements. Regarding current security controls, standard employee laptops utilize CrowdStrike endpoint protection and Active Directory authentication; however, the newly deployed VR headsets currently lack any endpoint security agents. The technology environment relies on a standard Cisco AnyConnect VPN for remote access. The primary assets that need protection in this environment are the mission critical Linux source code servers and the confidentiality of the employees' biometric data.

Problem Statement: *Describe the cybersecurity risk or vulnerability needing remediation*.

The primary cybersecurity problem I want to address is the severe privacy and security risk created by the new VR headsets. Currently, remote employees are connecting these unmanaged headsets directly to the internal corporate network using the standard VPN without any endpoint verification. This setup essentially bridges their home networks directly into the primary corporate IT environment, putting the headsets on the exact same network as standard employee laptops and the mission critical source code servers. Consequently, the headsets are transmitting highly sensitive, unencrypted biometric and spatial data through the unsegmented network, creating a massive threat to the confidentiality of our employees' personal data. Furthermore, because these remote devices lack standard enterprise endpoint protection or VPN posture checks, they represent a significant control weakness. The attack vector is clear: if a threat actor compromises an employee's poorly secured home network or the VR headset itself, they could easily use that trusted VPN connection as an opportunity to pivot into the main corporate network. From there, they could access our confidential source code servers or deploy ransomware, impacting our overall system confidentiality, integrity, and availability.

Cybersecurity Technical Control Solution

Proposed Control: *Describe the technical control solution.*

To solve this problem, I propose implementing a multi-layered security approach that focuses on Zero Trust principles and network segmentation to secure the unmanaged VR headsets (Rose et al., 2020). First, I will research and design a method to enforce VPN posture checks using a Mobile Device Management solution like Microsoft Intune. This means that before any headset is allowed to connect to the network, the system will verify that the device meets basic security requirements and is running the latest firmware. Second, I will propose the creation of a dedicated Virtual Local Area Network. By putting all 150 VR headsets on their own isolated network segment, we can ensure that even if a headset is compromised, the attacker cannot access the main corporate servers where sensitive data is stored. Finally, I will outline a plan to enforce strong encryption protocols, such as IPsec or TLS 1.3, for all spatial and biometric data while it travels from the remote headsets to the corporate network to align with modern privacy standards (XR Safety Initiative, 2020).

Scope: *Identify the systems, networks, and data impacted by the risk or vulnerability.*

The scope of my project will focus specifically on the 150 organizationally issued Meta Quest Pro VR headsets, the central Cisco AnyConnect VPN gateway that they connect through, and the internal Cisco routing rules needed to create the new /24 Virtual Local Area Network subnet. I will also include the protection of the spatial and biometric data generated during remote sessions. I will explicitly exclude the 450 standard employee Windows 11 laptops and the two primary Linux source code servers from this project scope. While those systems will directly benefit from the added network isolation, the

compensating control for those excluded devices is the existing Active Directory authentication and their currently installed CrowdStrike endpoint protection.

Objectives: *Describe what the technical control solution will do.*

The main objective of my technical control solution is to completely isolate the unmanaged VR traffic from our critical corporate systems. The desired end state of the environment is a fully segmented network architecture where all VR telemetry data is encrypted in transit and no unmanaged device can connect to the VPN without passing an automated posture check. By achieving this Zero Trust end state, I want to prevent any unauthorized network access and stop potential threat actors from pivoting through the VPN if a headset is compromised. Essentially, if an attacker manages to take over an employee's home VR headset, they could use that headset's existing VPN connection to sneak into the main corporate network and target our sensitive servers. By isolating the VR traffic into its own dedicated subnet, I can successfully cut off this lateral movement pathway. Another major objective is to protect employee privacy by ensuring the biometric and spatial telemetry data remains completely confidential.

Project Stakeholders

Describe the various people and groups (security team, IT, executive management, etc.) that need to be involved in the project.

To ensure this project is successful, I will need to collaborate with several key stakeholders. First, I will involve the IT Network Engineering team to help design and implement the dedicated VLAN and configure the VPN gateway. I will also need to work with the Information Security team to define the specific endpoint posture checks and encryption standards required for the headsets. Finally, I will keep Executive Management informed so they understand the privacy risks being mitigated and can approve the necessary resource allocation and project budget.

Implementation Plan

Outline how you will complete your project.

My plan will happen in three main phases over a six-week period. First, I will start with the Design Phase (Weeks 1 and 2), where I will work with the Information Security team to define the required posture checks and collaborate with the IT Network Engineering team to map out the new VLAN architecture. Next, in the Testing Phase (Weeks 3 and 4), the IT Network Engineering team will configure a pilot group of 10 headsets to connect to the new VLAN through the VPN. This will allow the Information Security team and me to verify that the posture checks and encryption work without disrupting the remote training sessions. Finally, in the Deployment Phase (Weeks 5 and 6), Executive Management will give the final approval to proceed. Following their approval, IT Network Engineering will roll out the new network rules and endpoint management policies to all 150 remote VR devices and monitor the network for any connectivity issues.

Sources

Acknowledge sources, using APA-formatted in-text citations and references, for content that is quoted, paraphrased, or summarized.

Rose, S., Borchert, O., Mitchell, S., & Connelly, S. (2020). Zero trust architecture (NIST Special Publication 800-207). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.SP.800-207
XR Safety Initiative. (2020). The XRSI privacy framework. https://xrsi.org/publication/the-xrsi-privacy-framework

Restricted Information Use

Potential use of restricted information (choose one): **Yes**☐ or **No** ☒

Potential use of human subjects (choose one): **Yes**☐ or **No** ☒

If you check "yes" to either statement, you must complete the "Restricted Information Authorization Form" and have a letter of authorization to use the information. The form (page 1) and the letter of authorization (page 2) must be completed and submitted with the topic approval form.

By submitting this form, you acknowledge that all information provided is accurate and that any changes to the topic, proposal, or goals must be discussed with your assigned instructor before continuing.

STUDENT SIGNATURE:__Kyle Joseph Picard_____

COURSE INSTRUCTOR NAME: John Jamison

COURSE INSTRUCTOR SIGNATURE: _____

COURSE INSTRUCTOR APPROVAL DATE: Click to enter a date.

10 March 2026